higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

27 November 2020 (X-paper)
09:00–12:00

NATED GO

This question paper consists of 6 pages.

296Q1E2027

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a new page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 $X + \overline{XY}$

A X
B 1
C $X + Y$
D $X + \bar{Y}$

1.1.2 $\overline{A + \overline{BC}}$

A BC
B ABC
C A
D $\bar{A}BC$

1.1.3 $A(A + B)$

A A
B 1
C AB
D B

1.1.4 $A + A\bar{B} + ABC$

A B
B A
C AB
D 1

1.1.5 $X(\overline{Y + Z})$

A Z
B XY
C $X\bar{Y}\bar{Z}$
D 0

(5 × 1) (5)

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1.2 Complete the following sentences by writing only the missing word or words next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 In the … system, only two symbols are used, 0 and 1.

1.2.2 The clock frequency is determined by the amount of … per second

1.2.3 The … consists of the arithmetic logic unit and the control unit.

1.2.4 The … was developed to reduce the number of bits by using exponents.

1.2.5 One method of proving the Boolean theorems is called … .

(5 × 1) (5)

1.3 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.3.1	The amount of logic gates in a single cell	A	mark period
1.3.2	Single output of a logic gate	B	weighted code
1.3.3	The period during which the pulse is active	C	bit
1.3.4	A code in which each bit position is allocated a fixed value	D	alphanumeric codes
1.3.5	Codes that can recognize numbers, letters and devised characters	E	packing density

(5 × 1) (5)

1.4 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

1.4.1 Five flip-flops are needed to count down from 18 to 0.

1.4.2 A decoder converts decimal to its binary equivalent.

1.4.3 A Johnson counter is a circulating register.

1.4.4 Flip-flops are mono-stable multi vibrators.

1.4.5 Medium-scale integration is logic circuits with 12 to 100 gates.

(5 × 1) (5)

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1.5 Choose ONE term from the list below for each of the following descriptions and write it next to the question number (1.5.1–1.5.5) in the ANSWER BOOK.

register; counter; byte; multiplexer; reset

1.5.1 A process which is used to alternatively send various signals through a single communication channel

1.5.2 A group of binary bits

1.5.3 A group of flip-flops that are used to store data in binary code

1.5.4 A digital sequential logic device that will go through a certain predefined state based on the application of the input pulses

1.5.5 An input that forces the Q output of a memory element to zero

(5 × 1) (5)

1.6 Choose the correct word from those given in brackets. Write only the word next to the question number (1.6.1–1.6.5) in the ANSWER BOOK.

1.6.1 In (asynchronous/synchronous) clocking flip-flops are clocked simultaneously.

1.6.2 A (decoder/encoder) converts binary to decimal equivalent.

1.6.3 Asynchronous flip-flop inputs are active (low/high).

1.6.4 A (digital/analogue) signal is a signal that is being used to represent data as a sequence of discrete values.

1.6.5 The Excess-3 code is an example of (weighted/unweighted) code.

(5 × 1) (5)

[30]

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QUESTION 2

2.1 Subtract by means of two's complement method and convert the final answer to the octal system.

2.1.1 $-25{,}75_{10} - 15{,}25_{10}$ (6)

2.1.2 $CF{,}5_{16} - AB{,}6_{16}$ (6)

2.2 Calculate the following in their binary form and convert the final answer to the decimal system.

2.2.1 $6A,7_{16} \div 5,6_8$ (6)

2.2.2 $B3,4_{16} \times 100,5_{10}$ (6)

2.3 Convert $56,7_{10}$ to the following BCD codes:

2.3.1 2421

2.3.2 8421

2.3.3 $XS\ 3$

(3 × 2) (6)
[30]

QUESTION 3

3.1 Simplify the following Boolean expressions:

3.1.1 $\bar{A}BC + A\bar{B}C + AB\bar{C} + B\bar{C}$ (4)

3.1.2 $AB\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}C$ (6)

3.2 Draw the logic circuit for the following Boolean expression:

$\bar{A}B + \bar{B}C + A(\overline{B + C}) = F$ (6)
[16]

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QUESTION 4

Draw a neat, fully labelled sketch of the following circuit diagrams:

4.1 4.1.1 Full adder (8)

4.2 4.1.2 4 lines to 1-line multiplexer system (6)
[14]

QUESTION 5

Make a neat sketch of the following circuit diagrams:

5.1 Module 5 counter (5)

5.2 3-Bit synchronous D-type up counter (5)
[10]

TOTAL: 100

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